



MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY
JAIPUR

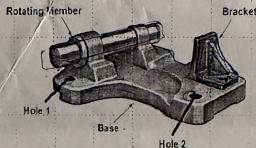
22MET101: Introduction to Mechanical Systems
Mid-term Exam – February 24, 2026
B. Tech. II Semester (A, B, C, D, E and F Batches)

MM: 30
Time: 90 min

Instructions: All the questions are compulsory. Marks and course objectives are indicated against each question. Draw sketches/labelled diagrams wherever necessary. Attempt the questions in the given sequence.

SECTION - A

- Q1. Give atleast two advantages and two disadvantages of road transportation over rail transportation. [2M, CO1]
- Q2. State the difference between a machine tool and a cutting tool. Give an example of both tools. [2M, CO2]
- Q3. An assembly is shown in the adjoining figure. It comprises three components, a rotating member, a base and a bracket with two mounting holes. Suggest suitable manufacturing processes to make each component of the assembly [2M, CO2]



- Q4. Which property of moulding sand facilitates easy escape of gases through it, and what type of casting defects can occur in its absence? With the help of which hand tool it can be artificially improved further? [2M, CO2]
- Q5. State one advantage and one drawback of subtractive manufacturing over additive manufacturing. Identify any one machine tool suitable for subtractive manufacturing of an engineered component and one suitable for additive manufacturing. [2M, CO2]

SECTION - B

- Q6. With a neat, labelled sketch, show the direction of tool feed for the execution of following lathe operations on a workpiece: (i) Taper turning, (ii) Knurling, (iii) Boring, (iv) Threading [5M, CO2]
- Q7. Draw a labelled diagram of a horizontal milling machine indicating various components and describe its working principle? [5M, CO2]
- Q8. State the operating principle of oxy-acetylene gas welding process. Show the welding equipment with a neat sketch and name the types of flames based on the relative proportions of fuel gas and oxidizing agent. [5M, CO2]
- Q9. Explain the significance of any five essential properties of moulding sand used for casting in foundry workshop. [5M, CO2]

Clamp 2m x 2m → 400 x 2m
400 x 100 → 400
400 x 100 → 400



Department of Mechanical Engineering

Mid-Term Examination

Course title: Introduction to Mechanical Systems

Course code: 22MET101

Programme/ Semester: B. Tech / II

Session: Jan – May 2025 (Even Semester)

Max. Marks: 30

Time Duration: 90 minutes

Date: 22/02/2025

- Note: 1. All the questions are compulsory. Marks are indicated against each question.
2. Use of scientific calculator and steam tables is permitted. Programmable/ graphical calculator is not allowed.
3. Draw neat labeled diagrams, sketches and graphs wherever required. All answers should be to the point.
4. Answer scripts with matching solution, partly or wholly, will be summarily rejected without prior information.
5. Attempt the questions in the given sequence. Answers of the numerical problems should be accompanied with S.I. units.
CO1: To understand the construction and working of a mobility system.
CO2: To ascertain the manufacturing processes appropriate for hardware development.
CO3: To select favourable engineering material(s) for the given application.
CO4: To analyze the efficiency of different thermal systems.
CO5: To select appropriate power transmission drive(s) for the given application.

Section A (5 x 2 = 10 marks)

1. Which property of the moulding sand facilitates easy escape of gases through it, and what type of casting defects can occur in its absence? With the help of which hand tool it can be artificially improved further? CO2
2. Distinguish between metal cutting and metal forming operations? Name atleast one lathe operation which involves metal cutting and metal forming, respectively. CO2
3. List any four modes of goods transportation in India. Also, highlight atleast one drawback of each mode. CO1
4. Match the following components of Centre Lathe with their respective functions. CO2
 - (a) Carriage (i) Supports the projected end of the work-piece
 - (b) Cross-slide (ii) Used for tool feed parallel to the bed ways
 - (c) Compound slide (iii) Used for tool feed inclined to the bed ways
 - (d) Tail stock (iv) Used for tool feed normal to the bed ways
5. For each of the following mechanical properties, name atleast one engineering material that represents the given property (a) Ductility (b) Malleability (c) Elasticity (d) Brittleness CO3

Section B (4 x 5 = 20 marks)

6. Identify the boundary of each thermodynamic system illustrated below and show it with a neat sketch. CO4
Also, state if the system is closed, open, or isolated



(a) Tyre



(b) Faucet



(c) Light Bulb



(d) Bucket



(e) DC Battery

7. Draw a neat labelled diagram of either a Horizontal Spindle Milling Machine or a Center Lathe, and list atleast four machining operations that can be executed using it. CO2
8. Adjacent figure shows a metallic cylindrical plate with a central hole. It will serve as a weight for resistance training in the gym. If the plate is to be made by performing sand casting in a foundry shop, list all the hand tools required and the step-by-step process (with a simple sketch depicting each step) for preparing the sand mould. CO2
9. Draw the electric circuit of a shielded metal arc welding equipment. What is the significance of "straight polarity" in an arc welding process? CO2

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
Department of Mechanical Engineering

First Year B.Tech (Batch: A,B,C,D,E,F)

Even Semester: 2023-24

Max Marks: 30

Mid Term Exam

Date : 28/02/2024

SUBJECT: Introduction to Mechanical Systems (22MET-101)

- Note: 1. All the questions are compulsory. Marks are indicated against each questions.
2. Use of Scientific calculator is permitted. Programmable/graphical calculator is not allowed.
3. Draw neat labelled diagrams, sketches and graphs whenever required. All answers should be to the point.
4. Answer script with matching solution, partly or wholly, will be summarily rejected without prior information.
5. Attempt the questions in the given sequence. Answer of the numerical problems should be accompanied with S.I units.

Section A (3 x 2 = 6 Marks)

- Que 1. List any four machining operations done on milling machine with neat diagram? (2)
- Que 2. Explain following mechanical properties of materials a) Strength b) Ductility (2)
c) Elasticity d) Hardness
- Que 3. Give the general classification of 3D printing and list basic steps of 3D printing? (2)

Section B (6 x 4 = 24 Marks)

- Que 4. Describe in brief brazing and soldering processes and their applications? (4)
- Que 5. What are the main characteristics which a good moulding sand should possess? (4)
- Que 6. What are the purposes of the following lathe parts? (4)
a) Tailstock b) Spindle c) Leadscrew d) Compound rest
- Que 7. Describe the functions of following tools used in foundry with neat diagram (4)
a) Strike - off bar b) Slick c) Rammer d) Vent rod
- Que 8. Explain the working of 2-stroke petrol engine with neat sketch. In what respect it differ from 4- stroke petrol engine? (4)
- Que 9. An air engine works on the following cycle: Air is taken in at atmospheric pressure of 1 bar and at temperature of 16°C and is compressed adiabatically, the pressure at the end of the stroke being 35 bar. Heat is taken in at constant pressure the expansion afterwards takes place adiabatically, the ratio of expansion being 5. The air exhausted at the end of the stroke, the heat is assumed to be rejected at constant volume. Find the ideal thermal efficiency. (4)
- Handwritten notes:*
For 1st part
1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30.



Department of Mechanical Engineering

Mid-Term Examination

Course title: Introduction to Mechanical Systems

Course code: 22MET101

Programme/ Semester: B. Tech. / I

Session: Aug – Dec 2023 (Odd Semester)

Max. Marks: 30

Time Duration: 90 minutes

Date: 25/09/2024

- Note: 1. All the questions are compulsory. Marks are indicated against each question.
2. Use of scientific calculator and steam tables is permitted. Programmable/ graphical calculator is not allowed.
3. Draw neat labeled diagrams, sketches and graphs wherever required. All answers should be to the point.
4. Answer scripts with matching solution, partly or wholly, will be summarily rejected without prior information.
5. Attempt the questions in the given sequence. Answers of the numerical problems should be accompanied with S.I. units.
- CO1: To understand the construction and working of a mobility system.
CO2: To ascertain the manufacturing processes appropriate for hardware development.
CO3: To select favourable engineering material(s) for the given application.
CO4: To analyze the efficiency of different thermal systems.
CO5: To select appropriate power transmission drive(s) for the given application.

Section A (5 x 2 = 10 marks)

1. List any four modes of goods transportation in India. Also, highlight atleast one drawback of each mode. CO1
2. For each of the following mechanical properties, name atleast one engineering material that represents the given property (a) Ductility (b) Malleability (c) Elasticity (d) Brittleness CO3
3. Name four essential elements of a Gas-welding equipment.
4. What is the purpose of a chuck on a Centre Lathe? Briefly explain the limitation of a three jaw chuck. CO2
5. List atleast four essential properties of moulding sand. CO2

Section B (4 x 5 = 20 marks)

6. In which workshop will you find the following: (a) Shovel (b) Lead screw (c) A.C. transformer (d) Lifter (e) File? Show each element with a neat diagram and briefly discuss the purpose of its use. CO2
7. For each of the following machining operations, name atleast one engineered component that was produced by performing the given operation (a) Chamfering (b) External threading (c) Knurling (d) Drilling (e) Step turning CO2
8. Draw a neat labelled diagram of either a Vertical Drilling Machine or a Horizontal Milling Machine, and briefly explain the function of each component. CO2
9. Adjacent figure depicts a machined component. If the raw material is a metallic rod with a square cross-section, list all of the machining operations in the sequence of their implementation, along with a neat sketch clearly depicting each stage of manufacturing.

